

UNIVERSAL HUMAN VALUES (UHV)

EXAM STUDY MATERIAL

10-Mark Topics • 2-Mark Topics • Visual flow diagrams for quick learning

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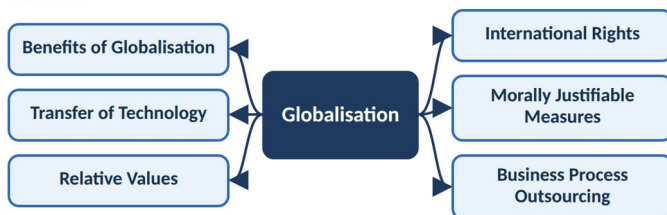
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PART A — 10-MARK QUESTIONS

1) GLOBALISATION — 10 MARKS

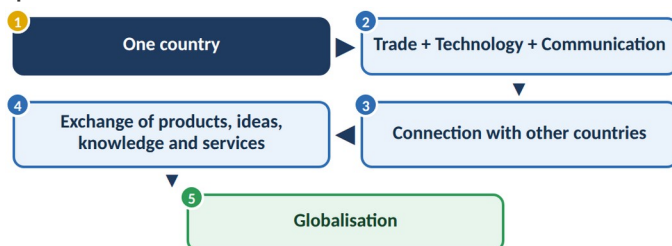
TOPIC MAP



Meaning of Globalisation

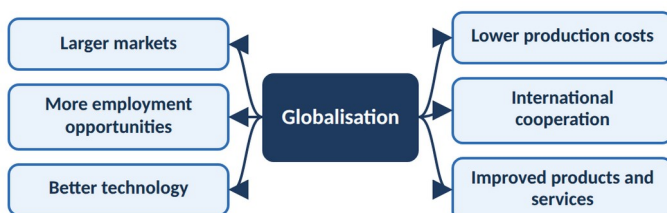
Globalisation is the process by which countries, businesses, people and technologies become increasingly connected with each other across the world.

Simple flow



1. Benefits of Globalisation

Globalisation provides several advantages:



Important benefits

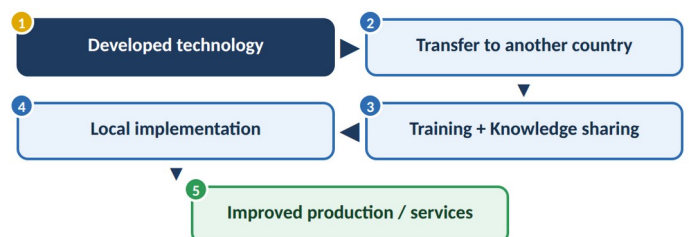
- **Larger market:** Companies can sell products to customers in many countries.

- **Employment:** International companies create jobs in different countries.
- **Technology development:** Countries can obtain and use advanced technologies.
- **Better quality:** Global competition encourages companies to improve quality.
- **Lower cost:** Production can be carried out where resources and labour are economical.
- **Knowledge sharing:** Countries can exchange scientific and technical knowledge.

2. Transfer of Technology

Transfer of technology means moving technical knowledge, machines, methods or expertise from one country or organisation to another.

Flow



Example

A company from Germany transfers advanced manufacturing technology to an Indian company.

The Indian company receives:

Technology + Equipment + Training + Technical knowledge

This helps the company produce better products.

Problems

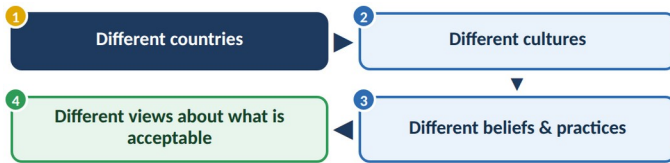
Technology transfer should be done carefully because:

- Old or unsuitable technology may be transferred.
- The receiving country may become dependent on another country.
- Environmental damage may occur.
- Intellectual property rights must be respected.

3. Relative Values

Relative values means that values, customs and moral expectations may differ between different societies and cultures.

Flow



For example, a business practice considered acceptable in one country may be considered inappropriate in another.

Therefore, engineers and companies working internationally must **respect cultural differences while maintaining basic ethical principles**.

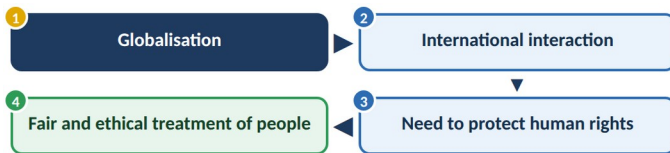
4. International Rights

International rights are basic rights that should be respected for people across countries.

Important rights include:

- Right to life and safety
- Right to equality
- Right to freedom
- Right to fair treatment
- Right to privacy
- Right to dignity
- Workers' rights

Simple idea



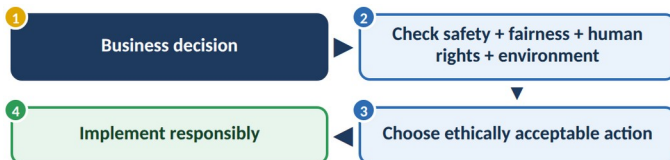
Engineers and organisations should ensure that their activities do not violate these rights.

5. Promoting Morally Justifiable Measures

Globalisation can create ethical problems such as exploitation of workers, environmental pollution and unfair business practices.

Therefore, organisations should take **morally justifiable measures**, meaning actions that are ethically acceptable and beneficial to people and society.

Ethical approach

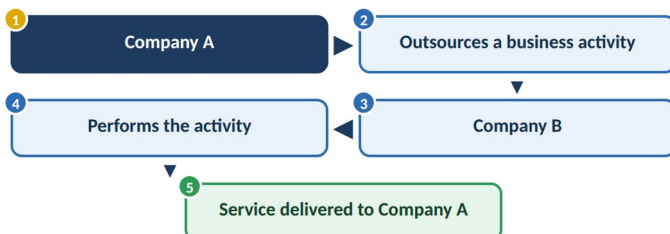


For example, a company should not reduce production costs by using unsafe working conditions.

6. Business Process Outsourcing (BPO)

BPO means giving certain business activities to an external company, often located in another country.

Flow



Examples:

- Customer support
- Data entry
- Accounting
- Technical support
- Call centres

Advantages

- Reduces operating costs
- Provides specialised skills
- Saves time

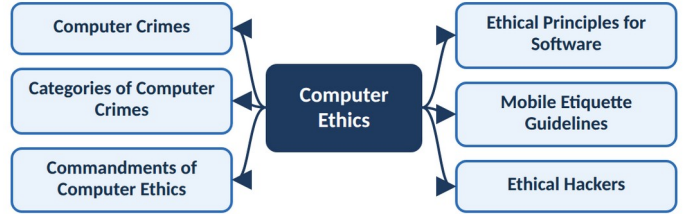
- Allows companies to focus on core activities

Ethical concerns

- Employee job security
- Data privacy
- Fair wages
- Working conditions
- Protection of confidential information

2) COMPUTER ETHICS — 10 MARKS

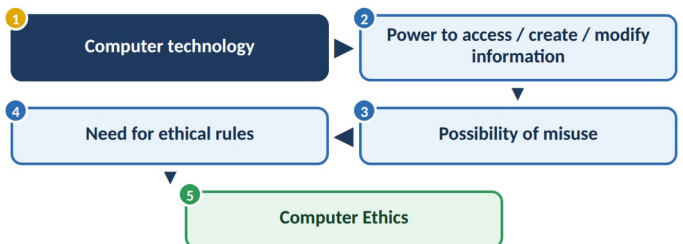
TOPIC MAP



Meaning of Computer Ethics

Computer ethics refers to the moral principles and rules that guide the responsible use of computers, software, networks and digital information.

Flow



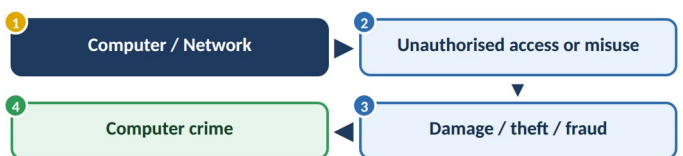
1. Computer Crimes

A **computer crime** is an illegal activity in which a computer, network or digital information is used as a target or as a tool.

Examples:

- Hacking
- Identity theft
- Online fraud
- Data theft
- Malware attacks
- Password theft
- Cyberstalking

Basic flow



2. Categories of Computer Crimes

Computer crimes can broadly be classified according to what is attacked.

A. Crimes against individuals

Target a particular person.

Examples:

- Identity theft
- Cyberstalking
- Online harassment
- Phishing

B. Crimes against property

Target digital or physical property.

Examples:

- Data theft
- Software piracy
- Intellectual property theft
- Malware causing damage

C. Crimes against organisations

Target companies or institutions.

Examples:

- Database attacks
- Ransomware
- Data breaches
- Denial-of-service attacks

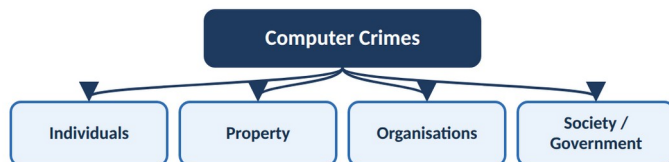
D. Crimes against society/government

Target large groups or public systems.

Examples:

- Cyberterrorism
- Attacks on government systems
- Spreading harmful information

Easy classification

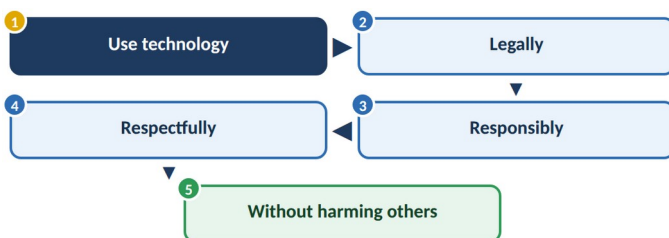


3. Commandments of Computer Ethics

The **Ten Commandments of Computer Ethics** provide basic rules for responsible computer use.

1. Do not use a computer to harm others.
2. Do not interfere with another person's computer work.
3. Do not access another person's files without permission.
4. Do not use a computer to steal.
5. Do not use a computer to spread false information.
6. Do not copy or use software without permission.
7. Do not use another person's computer resources without authorisation.
8. Do not take another person's intellectual work.
9. Think about the social consequences of the programs you create.
10. Use computers with consideration and respect for others.

Core idea



4. Ethical Principles for Software

Software engineers should develop and use software responsibly.

Important principles:

1. **Public interest**
Software should not unnecessarily harm users or society.
2. **Quality**
Software should be reliable, safe and properly tested.
3. **Privacy**
Users' personal information must be protected.
4. **Confidentiality**
Confidential information should not be disclosed without permission.
5. **Intellectual property**
Copyrights, patents and licenses must be respected.
6. **Honesty**
Developers should not intentionally provide false information about software.

Flow



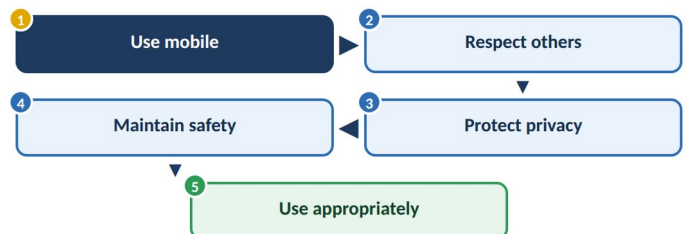
5. Mobile Etiquette Guidelines

Mobile etiquette means using mobile phones in a socially responsible and respectful manner.

Important guidelines:

- Keep the phone silent in classrooms, meetings and hospitals.
- Do not disturb others with loud conversations.
- Do not use phones while driving.
- Respect other people's privacy.
- Do not record or photograph people without permission.
- Avoid unnecessary calls in public places.
- Do not use phones during important discussions.
- Use mobile devices responsibly.

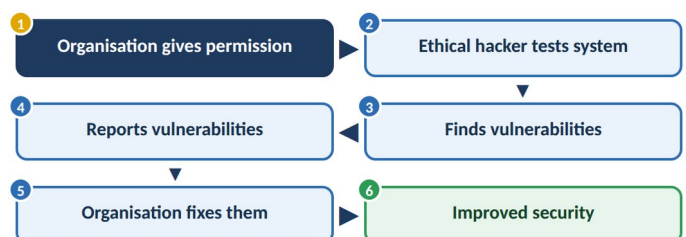
Simple idea



6. Ethical Hackers

An **ethical hacker** is a security professional who legally tests computer systems to find vulnerabilities before malicious hackers can exploit them.

Flow

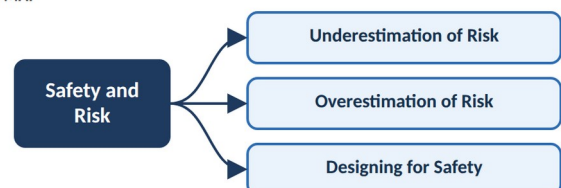


Ethical hacker vs malicious hacker

Ethical Hacker	Malicious Hacker
Has permission	No permission
Finds vulnerabilities	Exploits vulnerabilities
Protects systems	Damages/steals
Reports problems	Hides activities
Legal	Illegal

3) SAFETY AND RISK — 10 MARKS

TOPIC MAP



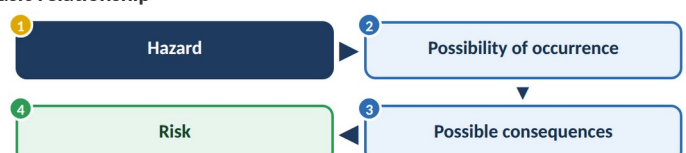
Meaning of Safety

Safety means protecting people, property and the environment from harm or unacceptable danger.

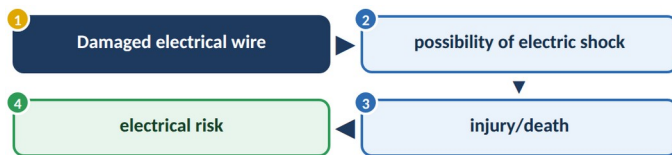
Meaning of Risk

Risk is the possibility that a hazard may cause harm, together with the seriousness of that harm.

Basic relationship



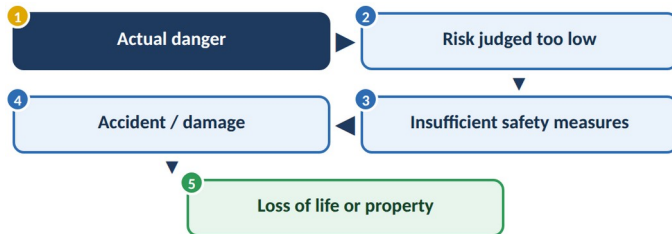
For example:



1. Underestimation of Risk

Underestimation of risk means judging a danger as less serious or less likely than it actually is.

Flow



Example

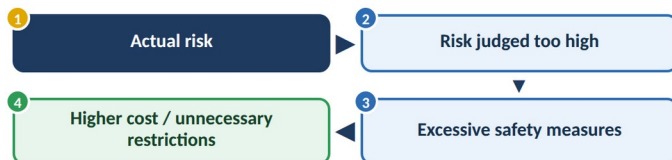
An engineer thinks a machine has only a very small chance of failure and does not install adequate protection.

If the machine fails, workers may be injured.

2. Overestimation of Risk

Overestimation of risk means judging a danger as greater than it actually is.

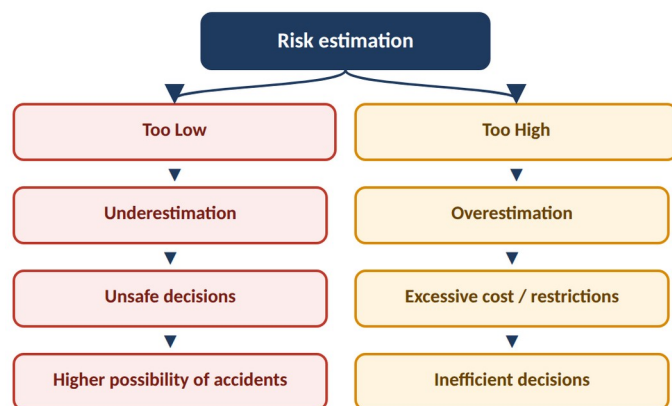
Flow



Example

If an engineer assumes that a very unlikely failure will definitely occur, they may spend excessive money on unnecessary protection.

Underestimation vs Overestimation

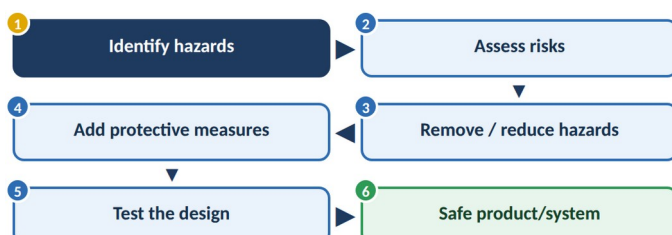


Therefore, engineers should aim for **realistic and scientifically supported risk assessment**.

3. Designing for Safety

Designing for safety means considering safety from the beginning of the engineering design process instead of trying to add safety after the product is completed.

Flow



Important methods

1. Eliminate the hazard

Best option: remove the dangerous condition completely.

2. Reduce the risk

If elimination is impossible, reduce the probability or severity of harm.

3. Protective devices

Use guards, alarms, emergency stops, insulation, etc.

4. Warning systems

Provide warnings where hazards cannot be completely removed.

5. Testing and maintenance

Regular testing ensures that safety features continue to work.

Important principle

Safety should be built into the design, not added as an afterthought.

4) ENGINEERING AS EXPERIMENTATION — 10 MARKS

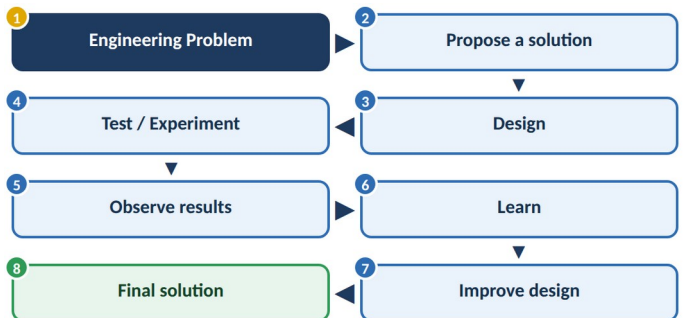
TOPIC MAP



Engineering is similar to experimentation because engineers often **design, test, observe results and improve their solutions**.

An engineering product may not behave exactly as predicted because real-world conditions are complex.

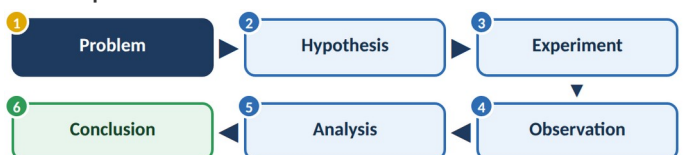
Main flow



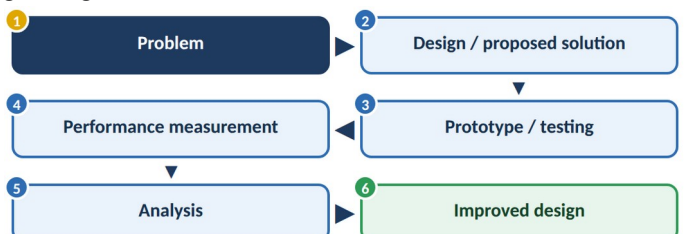
1. Similarities to Standard Experiments

Engineering activities have many similarities with scientific experiments.

Scientific experiment



Engineering



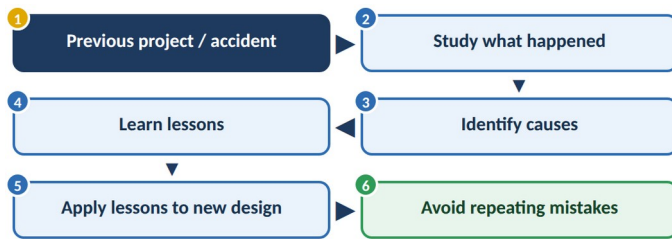
Both involve:

- Planning
- Testing
- Observation
- Data collection
- Analysis
- Learning from results

2. Learning from the Past

Engineers should study previous failures and successes before designing a new system.

Flow



Example

If an earlier bridge failed because of inadequate load calculations, engineers designing a new bridge should study that failure and improve the design calculations.

Past failure → Knowledge → Better future design

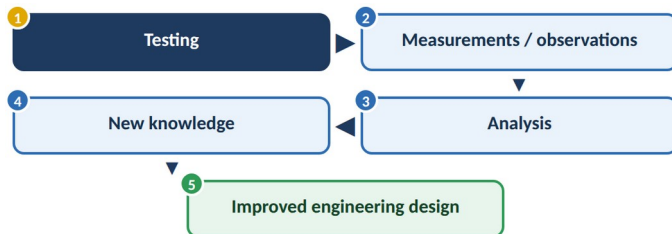
3. Knowledge Gained

Engineering experiments generate useful knowledge about the behaviour and performance of a system.

For example, testing a machine may reveal:

- Maximum load
- Temperature limits
- Failure points
- Efficiency
- Material behaviour
- Safety limits

Flow



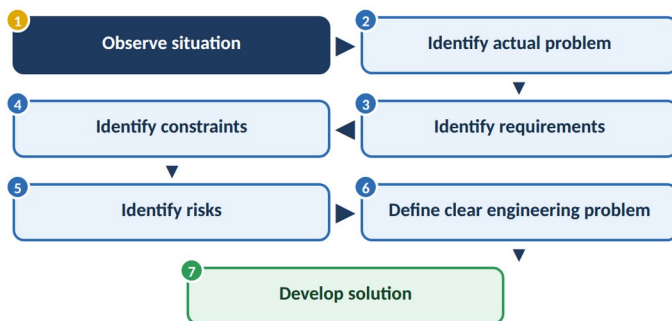
The knowledge gained should be properly documented and used in future engineering decisions.

4. Framing the Problem

Framing the problem means clearly defining what the actual engineering problem is before trying to solve it.

A poorly defined problem can lead to a wrong solution.

Flow



Example

Instead of saying:

✗ "The vehicle is unsafe."

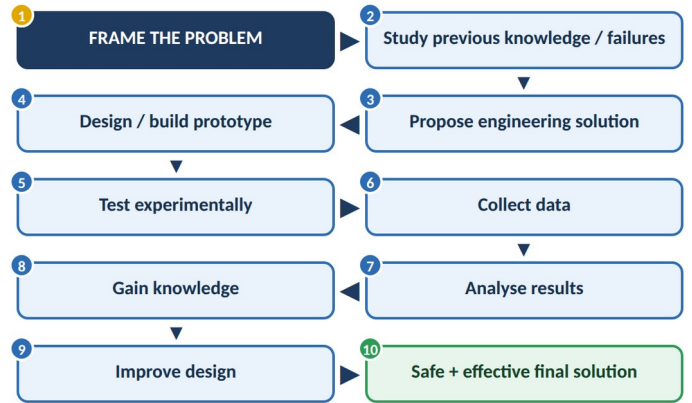
A better engineering problem statement is:

✓ "The vehicle's braking distance is too long at high speeds and must be reduced while maintaining cost and passenger comfort."

Now the engineer knows exactly what needs to be improved and what constraints must be considered.

Complete Engineering Experimentation Concept

You can remember the entire topic through this single flow:



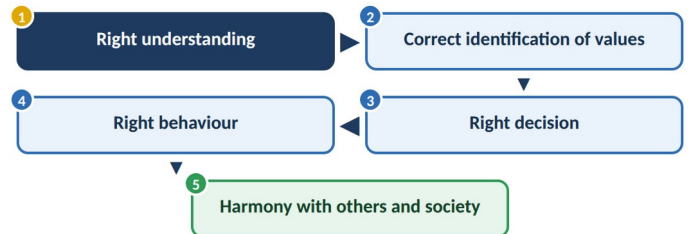
PART B — 2-MARK QUESTIONS

1) DEFINITIVENESS OF ETHICAL HUMAN CONDUCT — 5 MARKS

Ethical human conduct means behaving in a way that is morally right, responsible and beneficial to oneself and others.

Definitiveness means that ethical behaviour should be **clear, consistent and based on correct understanding**, rather than changing according to convenience or personal benefit.

Basic concept



Main points

- Ethical conduct should be based on **right understanding and universal human values**.
- A person should distinguish between **right and wrong** before taking decisions.
- Ethical behaviour should be **consistent**, even when there is pressure or temptation.
- It should promote **mutual happiness, trust and respect**.
- A person should consider the **effect of their actions on others and society**.
- Ethical conduct should not depend only on fear of punishment or personal gain.

Example

An engineer discovers a safety defect in a product.

Personal benefit: Hide the defect → earn profit

Ethical conduct: Report the defect → protect users

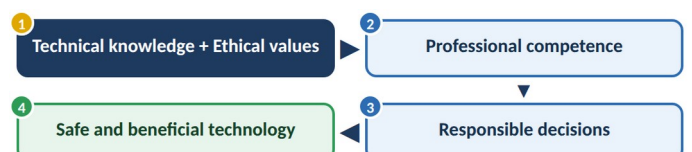
Therefore, ethical human conduct means **choosing the right action based on values and responsibility, even when it is difficult**.

2) COMPETENCE IN PROFESSIONAL ETHICS — 5 MARKS

Professional ethical competence is the ability of a professional to combine **technical knowledge with ethical understanding** and make responsible decisions.

An engineer should not only know **how to build something**, but also understand **whether it is safe, fair and beneficial**.

Flow



Main requirements

An ethically competent engineer should:

- Have sufficient **technical knowledge and skills**.
- Understand **professional codes of ethics**.
- Give importance to **public safety and welfare**.
- Be honest about technical limitations and risks.

- Protect confidential information.
- Respect intellectual property.
- Avoid discrimination and conflicts of interest.
- Take responsibility for professional decisions.
- Continuously improve knowledge and skills.

Example

An engineer is asked to approve a product even though testing shows a serious safety problem.

A technically competent but unethical engineer may approve it.

A **professionally ethically competent engineer** will:

Identify risk → refuse unsafe approval → report the problem → protect users.

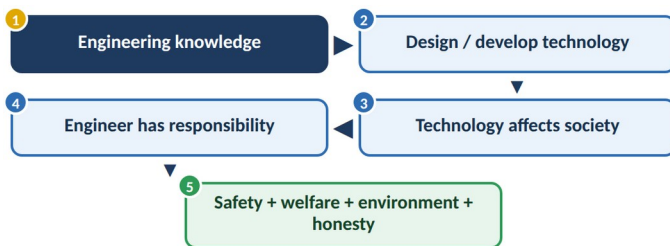
Thus, professional competence requires **both technical excellence and ethical responsibility**.

3) ROLE AND RESPONSIBILITIES OF ENGINEERS — 5 MARKS

Engineers use scientific and technical knowledge to design and develop products, systems and technologies that affect society.

Therefore, their responsibility is not only towards their employer but also towards **public safety, society and the environment**.

Flow



Major responsibilities

1. Protect public safety

Engineers must ensure that their designs do not unnecessarily endanger people.

2. Maintain professional competence

They should continuously update their technical knowledge and skills.

3. Be honest

Engineers should provide accurate information and should not hide important risks or defects.

4. Protect the environment

Engineering activities should minimise pollution, waste and environmental damage.

5. Follow laws and ethical standards

Engineers must comply with applicable laws, regulations and professional codes.

6. Maintain confidentiality

Confidential information obtained through professional work should be protected.

7. Report unsafe practices

Engineers should raise concerns when they identify serious safety or ethical problems.

8. Consider society

Engineers should consider the social consequences of their technologies.

Example

If an engineer discovers that a bridge design has a serious structural weakness, the engineer must **report it and ensure corrective action**, even if it causes delay or financial loss.

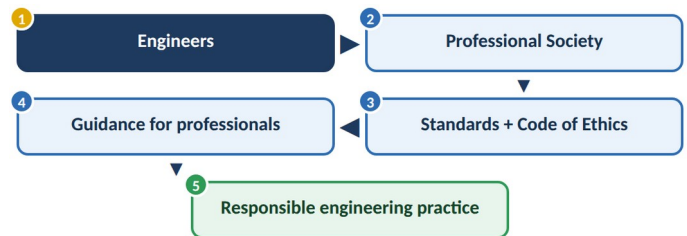
4) PROFESSIONAL SOCIETIES AND THEIR CODES OF ETHICS — 5 MARKS

Professional Societies

A **professional society** is an organisation formed by people belonging to a particular profession to promote professional knowledge, standards and responsible practice.

Examples include engineering professional organisations that establish standards and ethical guidelines for engineers.

Flow



Functions of Professional Societies

- Promote professional knowledge and education.
- Organise conferences, seminars and technical activities.
- Develop professional standards.
- Encourage ethical behaviour.
- Provide professional development opportunities.
- Protect the reputation and integrity of the profession.
- Help professionals understand their responsibilities.

Codes of Ethics

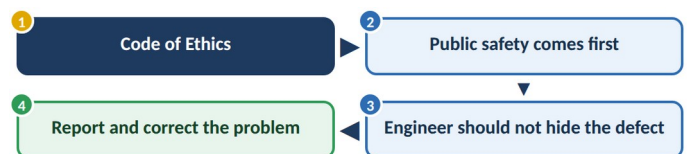
A **code of ethics** is a set of principles and rules that guides professionals in making ethical decisions.

Important principles include:

- **Protect public safety and welfare.**
- Be **honest and trustworthy**.
- Maintain professional **competence**.
- Avoid **conflicts of interest**.
- Respect **confidentiality**.
- Respect intellectual property.
- Follow laws and professional standards.
- Act responsibly towards society and the environment.

Example

If an engineer's company asks them to hide a dangerous defect:



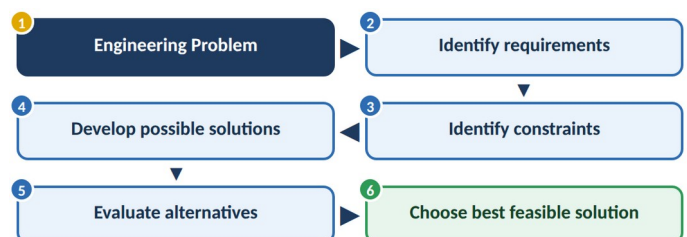
Thus, professional societies and their codes of ethics help engineers **maintain professional standards and make responsible decisions**.

5) CONSTRAINTS IN ENGINEERING — 5 MARKS

Engineering constraints are the limitations or conditions that engineers must consider while designing and developing a product or system.

An engineer cannot design something based only on what is technically possible. The design must also satisfy practical limitations.

Flow



Major Engineering Constraints

1. Cost

The solution must usually remain within the available budget.

2. Time

The project must be completed within the required deadline.

3. Materials

The availability, strength and properties of materials limit the design.

4. Technology

Available tools, machines and technical knowledge can restrict what can be developed.

5. Safety

The design must not create unacceptable risks to users or workers.

6. Environmental constraints

The design should minimise pollution, waste and environmental damage.

7. Legal constraints

Engineers must follow laws, regulations and standards.

8. Social and ethical constraints

The solution should respect human rights, social values and ethical principles.

Example

Suppose an engineer has to design an electric vehicle.

The design must satisfy:

Low cost + adequate battery range + safety + available materials + environmental requirements + legal standards

The engineer therefore cannot simply choose the most powerful battery. The solution must satisfy **multiple constraints simultaneously**.

Important concept

Best engineering solution ≠ Most powerful solution

It is the solution that provides the **best balance between performance, cost, safety, time, environmental impact and other requirements**.